

Conceptual Framework for Estimating LLIN Impact

OpenMalaria simulation model



Ghana demography
Seasonality
Vector species
Health system access

Transmission intensity scenarios (*EIR* sweep)

Low *EIR*



Moderate *EIR*



High *EIR*



LLIN usage scenarios

Proportion of population using LLIN (0% → 80%)



LLIN Users

Rest of the population at each usage level

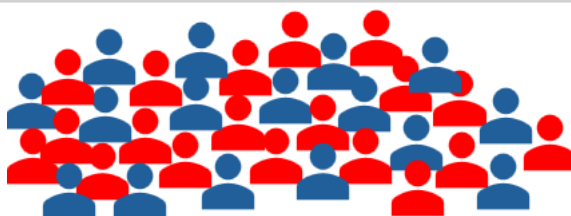


Non-users

Counterfactual baseline

- LLIN usage=0
- Non-user prevalence

Population Stratification (0 – 100 years)



Users and non-users coexist

Analysis outputs

Direct protection

Prevalence among LLIN users
(vs counterfactual)

Indirect protection

Prevalence among non-users
(vs counterfactual)

Overall protection

Population-weighted prevalence reduction
(vs counterfactual)

Community effect

Additional benefit beyond direct protection (absolute difference between direct and overall protection)